LIPSCHITZ NOISE IN TWO-POINT ZEROTH-ORDER NONSMOOTH CONVEX OPTIMIZATION: THE ADMISSIBLE SLOPE IS DIMENSION-FREE

Anonymous authors
Paper under double-blind review

ABSTRACT

We study two-point zeroth-order methods for nonsmooth convex optimization when function values are corrupted by deterministic noise that is neither random nor fully adversarial: the noise is the sum of an M-Lipschitz function of the query point and a component with oscillation at most Δ_0, while its amplitude is unrestricted. This class contains bounded adversarial noise, Lipschitz systematic errors and floating-point rounding. For an M_f-Lipschitz objective on a set of diameter D, we show that projected two-point SGD with ℓ_2-ball smoothing reaches accuracy ε with the optimal $O(d(M_f + M)^2 D^2/\varepsilon^2)$ oracle calls whenever $M \lesssim \varepsilon/D$ and $\Delta_0 \lesssim \varepsilon^2/(\sqrt{d} M_f D)$, and that no algorithm succeeds when $M > 2\varepsilon/D$. Thus the admissible Lipschitz slope is $\Theta(\varepsilon/D)$, independent of the dimension; prior analyses of exactly this setting obtained $\varepsilon/(\sqrt{d}D)$, and the $\sqrt{d}$ was conjectured to be necessary. The key step is to average before bounding: by Stokes' formula the noise contributes the gradient of the smoothed noise, whose norm never exceeds M. We show that a $\sqrt{d}M$ bias does arise if the noise may depend on the method's current random direction. Experiments confirm the dimension-free threshold, and on fp16/bf16 rounding reveal a "dead zone" with threshold $\gamma_{\mathrm{crit}} \approx 0.2\sqrt{d}\, q/M_f$ for the smoothing radius, where q is the rounding step.

1 INTRODUCTION

Zeroth-order (gradient-free) methods minimize a function using only its values. They are the tool of choice when gradients are unavailable or unreliable, from simulation-based design to memory-efficient fine-tuning of large language models with forward passes only (Malladi et al., 2023). When the function is computed by a program, exact values are rarely the bottleneck: backpropagation gives the gradient at a constant-factor cost. What makes the zeroth-order setting genuinely necessary is *inexactness* of the values, and the quality of a method is therefore measured by three numbers (Gasnikov et al., 2022a): the number of oracle calls T, the number of sequential rounds N, and the largest admissible noise level for which accuracy ε is still attainable.

The standard construction for a nonsmooth convex M_f-Lipschitz objective f (Gasnikov et al., 2022b) smooths f over a Euclidean ball of radius γ, estimates the gradient of the smoothed function with the two-point estimator of Shamir (2017),

$$g(x, e) = \frac{d}{2\gamma}\big(\tilde{f}(x + \gamma e) - \tilde{f}(x - \gamma e)\big)e, \qquad e \sim \mathrm{Unif}(S^{d-1}), \tag{1}$$

and feeds it into an optimal first-order method. Here $\tilde{f} = f + \delta$ are the observed values. For *bounded adversarial* noise $|\delta| \le \Delta$ the noise creates a bias of order $\sqrt{d}\Delta/\gamma$, which limits the admissible level to $\Delta \lesssim \varepsilon^2/(\sqrt{d} M_f D)$; this is optimal for not too small ε (Risteski & Li, 2016). The threshold is tiny because an adversary may place jumps exactly where the method probes.

Many systematic errors are not like that: rounding, discretization and model errors change slowly with the input. Gasnikov et al. (2016) already modeled such errors as a Lipschitz function plus a part that cancels in the two-point difference. For the nonsmooth two-point setting with an M-Lipschitz noise δ, Dvinskikh et al. (2022) and Lobanov & Gasnikov (2024) obtained the admissible

Table 1: Admissible noise for nonsmooth convex two-point zeroth-order optimization (M_f-Lipschitz f, diameter D; constants and logarithms omitted).

Noise model	Admissible level	Reference
bounded, $\lvert\delta\rvert \le \Delta$	$\Delta \lesssim \varepsilon^2/(\sqrt{d}M_f D)$; tight for $\varepsilon \gtrsim M_f D/\sqrt{d}$	Gasnikov et al. (2022b); Risteski & Li (2016)
M-Lipschitz	$M \lesssim \varepsilon/(\sqrt{d}D)$	Dvinskikh et al. (2022); Lobanov & Gasnikov (2024)
M-Lipschitz, bounded Hessian of f	$M = \Theta(\varepsilon/D)$	Loginov et al. (2026)
$\mathcal{N}(M, \Delta_0)$, nonsmooth f	$M = \Theta(\varepsilon/D),\ \Delta_0 \lesssim \varepsilon^2/(\sqrt{d}M_f D)$	**this paper**

level $M \lesssim \varepsilon/(\sqrt{d}R)$, and Dvinskikh et al. (2022, Appendix B) conjectured that this $\sqrt{d}$ is necessary. The recent work of Loginov et al. (2026) proves the dimension-free threshold $\Theta(\varepsilon/R)$ for objectives with a bounded Hessian, but still quotes $\varepsilon/(R\sqrt{d})$ for the nonsmooth case.

Contributions.

- We introduce the noise class $\mathcal{N}(M, \Delta_0)$: $|\delta(x) - \delta(y)| \le \Delta_0 + M\|x - y\|$ (Section 3). It interpolates between Lipschitz noise ($\Delta_0 = 0$) and bounded noise ($M = 0$), contains rounding, and puts no restriction on the amplitude.
- **Upper bound** (Theorem 1). Projected two-point SGD with explicit parameters reaches $\mathbb{E}f(\bar{x}_N) - f^* \le \varepsilon$ with $O(d(M_f + M)^2 D^2/\varepsilon^2)$ calls if $M \le \varepsilon/(6D)$ and $\Delta_0 \le \varepsilon^2/(9\sqrt{d}M_f D)$. The call complexity is optimal (Duchi et al., 2015).
- **Lower bound** (Theorem 2). If $M > 2\varepsilon/D$, no algorithm with any number of queries is ε-accurate. Hence the admissible slope is $\Theta(\varepsilon/D)$ in the nonsmooth case, improving the known $\varepsilon/(\sqrt{d}D)$ by $\sqrt{d}$ and refuting the conjecture of Dvinskikh et al. (2022) for globally Lipschitz noise.
- **Where $\sqrt{d}$ comes from** (Proposition 1). The analysis extends to noise chosen adaptively from the history, but if the noise may depend on the current random direction, a bias of $\sqrt{2d/\pi}\, M$ is attainable.
- **Experiments** (Section 5) confirm the dimension-independence of the Lipschitz threshold and the $1/\sqrt{d}$ scaling of the bounded one, and show that for real fp16/bf16 rounding the dominant failure is a "dead zone" with an empirical threshold $\gamma_{\mathrm{crit}} \approx 0.2\sqrt{d}\, q/M_f$.

We stress that the ingredients are known. The dimension-free bias bound is essentially the observation of Gasnikov et al. (2016, Sec. 4), made there with a suboptimal n^2 call complexity; the lower bound is in the spirit of Loginov et al. (2026, Prop. E.7); and a direction-aware noise construction appears in Loginov et al. (2026, Remark D.5) for a different estimator. Our contribution is to put them together in the nonsmooth two-point setting with the optimal call complexity, to identify precisely the step where previous analyses lose $\sqrt{d}$, and to extend the result to the mixed class $\mathcal{N}(M, \Delta_0)$.

2 Related work

Smoothing and two-point estimators. Randomized smoothing over a ball goes back to Nemirovsky & Yudin (1983) and Flaxman et al. (2005); Gaussian smoothing was analyzed by Nesterov & Spokoiny (2017). Duchi et al. (2015) established the optimal d-dependence of two-point methods, and Shamir (2017) showed that the estimator equation 1 has second moment $O(dM_f^2)$ independent of γ. Gasnikov et al. (2022b) combined this estimator with batched accelerated methods (Lan, 2012) to obtain methods with optimal calls and few rounds; see the survey Gasnikov et al. (2022a).

Inexact values. Bounded deterministic noise and the maximum admissible noise level were studied by Bayandina et al. (2018); Gasnikov et al. (2022b); Pasechnyuk-Vilensky et al. (2023), with lower bounds from approximately convex optimization (Risteski & Li, 2016). Inexact first-order

oracles were analyzed by Devolder et al. (2014). Lipschitz noise appeared in Gasnikov et al. (2016) (with a part that cancels in the two-point difference), in Dvinskikh et al. (2022) for nonsmooth saddle-point problems, and in Lobanov & Gasnikov (2024). For highly smooth functions, Loginov et al. (2026) determine the admissible Lipschitz systematic error up to constants.

3 SETTING

Let $Q \subset \mathbb{R}^d$, $d \geq 2$, be a closed convex set of diameter D, and let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and M_f-Lipschitz (queries may leave Q by at most γ). Let $f^* = \min_Q f$. The oracle returns $\tilde{f}(x) = f(x) + \delta(x)$, where δ is a Borel function fixed before the run.

Definition 1. The noise δ belongs to $\mathcal{N}(M, \Delta_0)$ if $|\delta(x)-\delta(y)| \leq \Delta_0 + M\|x-y\|$ for all $x, y \in \mathbb{R}^d$.

Bounded noise $|\delta| \leq \Delta$ lies in $\mathcal{N}(0, 2\Delta)$; M-Lipschitz noise of any amplitude lies in $\mathcal{N}(M, 0)$; rounding of the values to a grid of step q lies in $\mathcal{N}(0, q)$. The model of Gasnikov et al. (2016) is the case $\Delta_0 = 0$ after cancellation in the difference.

Method. Start at $x_0 \in Q$, draw i.i.d. $e_k \sim \mathrm{Unif}(S^{d-1})$ and set

$$g_k = \frac{d}{2\gamma}\big(\tilde{f}(x_k+\gamma e_k) - \tilde{f}(x_k-\gamma e_k)\big)e_k, \qquad x_{k+1} = \Pi_Q(x_k - h g_k), \qquad \bar{x}_N = \frac{1}{N}\sum_{k=0}^{N-1} x_k. \tag{2}$$

Each iteration uses two oracle calls. Let $f_\gamma(x) = \mathbb{E}_u f(x + \gamma u)$, $u \sim \mathrm{Unif}(B^d)$, be the ball smoothing of f; recall that $f \leq f_\gamma \leq f + \gamma M_f$ and that f_γ is convex and differentiable.

4 MAIN RESULTS

4.1 BIAS AND SECOND MOMENT OF THE ESTIMATOR

The first lemma separates the slope from the jumps.

Lemma 1 (decomposition)**.** *Let $\delta \in \mathcal{N}(M, \Delta_0)$ and $\delta_L(x) = \inf_y\big[\delta(y) + M\|x - y\|\big]$. Then δ_L is M-Lipschitz and $0 \leq \delta - \delta_L \leq \Delta_0$. Conversely, an M-Lipschitz function plus a function with values in $[0, \Delta_0]$ lies in $\mathcal{N}(M, \Delta_0)$.*

Lemma 2 (bias and second moment)**.** *Let $\delta \in \mathcal{N}(M, \Delta_0)$ and let g be the estimator equation 1 at a fixed x. Then*

$$\mathbb{E}_e g = \nabla f_\gamma(x) + b(x), \qquad \|b(x)\| \leq M + \frac{\sqrt{d}\,\Delta_0}{2\gamma}, \qquad \mathbb{E}_e\|g\|^2 \leq 4d(M_f + M)^2 + \frac{d^2\Delta_0^2}{2\gamma^2}. \tag{3}$$

Why there is no $\sqrt{d}$ in front of M. Write $\delta = \delta_L + \delta_B$ by Lemma 1. The key identity is the Stokes (divergence) formula for the ball (Flaxman et al., 2005): for any continuous h,

$$\frac{d}{2\gamma}\mathbb{E}_e\big[(h(x+\gamma e) - h(x-\gamma e))e\big] = \nabla h_\gamma(x). \tag{4}$$

Applied to $h = \delta_L$, the slope part of the noise contributes *exactly* the gradient of the smoothed noise, and $\|\nabla(\delta_L)_\gamma\| \leq M$ because smoothing preserves the Lipschitz constant. Previous analyses (Dvinskikh et al., 2022; Lobanov & Gasnikov, 2024) bounded the integrand first, $|\delta(x+\gamma e) - \delta(x-\gamma e)| \leq 2\gamma M$, and only then averaged:

$$\Big\langle \tfrac{d}{2\gamma}\mathbb{E}_e\big[(\delta(x+\gamma e) - \delta(x-\gamma e))e\big], r\Big\rangle \leq dM\,\mathbb{E}|\langle e, r\rangle| \approx \sqrt{2d/\pi}\,M\|r\|.$$

Bounding before averaging discards the sign cancellations that make equation 4 small; the resulting $\sqrt{d}$ is attained only by noise that knows e (Proposition 1). For the jump part no cancellation is available, and the bound $\frac{d}{2\gamma}\Delta_0\mathbb{E}|\langle e, r\rangle| \leq \frac{\sqrt{d}\Delta_0}{2\gamma}\|r\|$ is the usual one. The second moment follows from the Poincaré inequality on the sphere (Ledoux, 2001), as for exact values (Shamir, 2017). Full proofs are in Appendix A.

4.2 UPPER BOUND

We use the standard guarantee for projected SGD with a biased gradient (Lemma 3 in Appendix A): if $\mathbb{E}[g_k \mid \mathcal{F}_k] = \nabla F(x_k) + b_k$ with $\|b_k\| \le B$ and $\mathbb{E}[\|g_k\|^2 \mid \mathcal{F}_k] \le G^2$, then $\mathbb{E}F(\bar{x}_N) - \min_Q F \le D^2/(2hN) + hG^2/2 + BD$. The bias contributes a floor BD that does not grow with N.

Theorem 1 (upper bound)**.** *Let $\varepsilon \le M_f D$ and $\delta \in \mathcal{N}(M, \Delta_0)$ with*

$$M \le \frac{\varepsilon}{6D}, \qquad \Delta_0 \le \frac{\varepsilon^2}{9\sqrt{d}\, M_f D}. \tag{5}$$

Run equation 2 with $\gamma = \varepsilon/(3M_f)$, $G^2 = 5d(M_f + M)^2$, $N = \lceil 45\, d(M_f + M)^2 D^2/\varepsilon^2 \rceil$ and $h = D/(G\sqrt{N})$. Then $\mathbb{E}f(\bar{x}_N) - f^* \le \varepsilon$ after $T = 2N = O(d(M_f + M)^2 D^2/\varepsilon^2)$ oracle calls. The amplitude of the noise does not enter.

Proof sketch. By Lemma 2 and equation 5, $\|b\| \le \varepsilon/(6D) + \varepsilon/(6D) = \varepsilon/(3D)$, and the jump term in the second moment is $d\varepsilon^2/(18D^2) \le d(M_f + M)^2$, so $G^2 = 5d(M_f + M)^2$ bounds the second moment. The SGD lemma applied to $F = f_\gamma$ gives $\mathbb{E}f_\gamma(\bar{x}_N) - \min_Q f_\gamma \le DG/\sqrt{N} + BD \le \varepsilon/3 + \varepsilon/3$. Finally $f \le f_\gamma$ and $\min_Q f_\gamma \le f^* + \gamma M_f = f^* + \varepsilon/3$. □

4.3 LOWER BOUND AND TIGHTNESS

Theorem 2 (lower bound)**.** *Let Q be a ball of diameter D, $\varepsilon < M_f D/2$ and $M > 2\varepsilon/D$. For every algorithm (randomized, adaptive, with any finite number of queries anywhere in $\mathbb{R}^d$ and output $\hat{x} \in Q$) there exist a convex M_f-Lipschitz f and $\delta \in \mathcal{N}(M, 0)$ with $\mathbb{E}f(\hat{x}) - f^* > \varepsilon$.*

Proof. Pick $a \in (2\varepsilon/D, \min\{M, M_f\}]$ and a unit vector u, and set $f_\pm(x) = \pm a\langle u, x\rangle$, $\delta_\pm = -f_\pm$. Both noises are a-Lipschitz, and the oracle answers 0 to every query in both cases, so the law of $\hat{x}$ is the same. Since $\min_Q f_\pm = -aD/2$ (for Q centered at the origin), $\mathbb{E}[f_+(\hat{x}) - f_+^*] + \mathbb{E}[f_-(\hat{x}) - f_-^*] = aD > 2\varepsilon$. □

The construction is the elementary core of Loginov et al. (2026, Prop. E.7), which uses quadratic objectives and applies to any class containing them. Together, Theorems 1 and 2 give:

Corollary 1. *In the nonsmooth two-point setting the admissible Lipschitz slope is $\Theta(\varepsilon/D)$, independent of d and of the noise amplitude (the constants differ by a factor 12). For the jump part, $\mathcal{N}(0, 2\Delta)$ contains bounded noise, for which the lower bound of Risteski & Li (2016) gives $\Delta_{\max} \lesssim \max\{\varepsilon^2/(\sqrt{d}M_f D), \varepsilon/d\}$ up to logarithms; hence for $\varepsilon \gtrsim M_f D/\sqrt{d}$ condition equation 5 is tight in both parameters.*

A smoothed step ("ramp") $\psi(x) = \Delta\, \mathrm{clip}(M\langle u, x\rangle/\Delta, -1, 1)$ lies in both $\mathcal{N}(M, 0)$ and $\mathcal{N}(0, 2\Delta)$, so Theorem 1 applies if *either* $M \le \varepsilon/(6D)$ *or* $2\Delta \le \varepsilon^2/(9\sqrt{d}M_f D)$: the harmless region in the (amplitude, slope) plane is L-shaped (Figure 1).

4.4 ADAPTIVE NOISE AND THE ORIGIN OF $\sqrt{d}$

Proposition 1. *(a) Let the noise $\delta_k \in \mathcal{N}(M, \Delta_0)$ used at step k be chosen from the history $(e_0, \dots, e_{k-1})$ (jointly measurably), but not from e_k. Then Theorem 1 still holds. (b) If the noise may depend on e_k, then already for $\Delta_0 = 0$ the bias can equal $dM\,\mathbb{E}|\langle e, v\rangle|\, v \approx \sqrt{2d/\pi}\, M\, v$ for a unit v: take $\delta_k(x) = M\,\mathrm{sign}\langle e_k, v\rangle\, \langle e_k, x - x_k\rangle$.*

Part (b) shows that the $\sqrt{d}$ of previous analyses is exactly the price of a noise that sees the current direction. Since the query points $x_k \pm \gamma e_k$ reveal e_k, the requirement in (a) means that δ_k is fixed before the queries of step k.

5 EXPERIMENTS

All experiments use the method equation 2 or exact one-dimensional quadrature of the bias, fixed random seeds, and run in under three minutes on one CPU core; the code and 24 unit tests that numerically check every lemma are in the supplementary material. Details are in Appendix B.

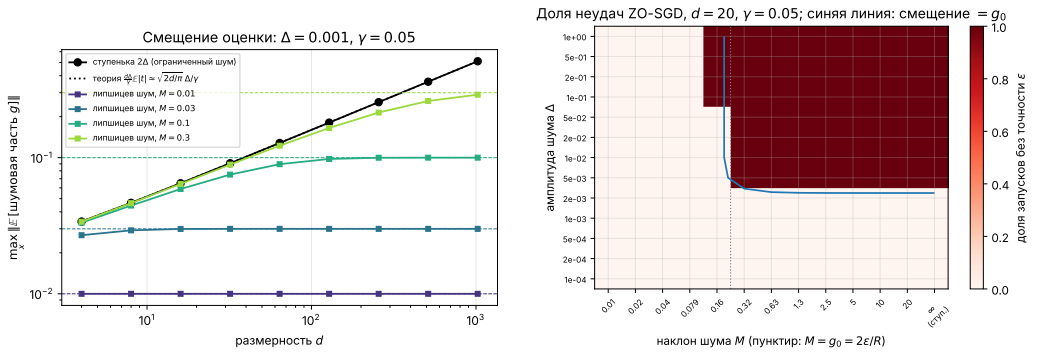


Figure 1: **Left:** maximal bias of the two-point estimator, computed by exact quadrature, for a step of height 2Δ (grows as $\sqrt{2d/\pi}\,\Delta/\gamma$) and for ramps of slope M (never exceeds M; equals M for $d \geq 128$). **Right:** success of the method against a noise "wall" of amplitude Δ and slope M, $d = 20$; the prediction "the method stops iff the bias exceeds the gradient of the problem" agrees with the experiment in 97.6% of the cells, and the safe region is L-shaped.

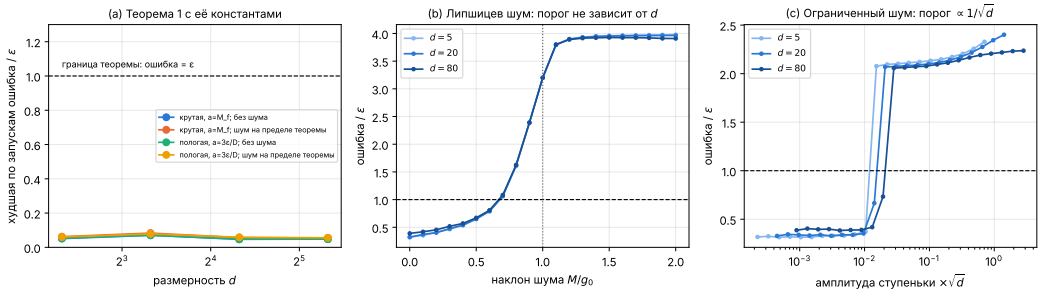


Figure 2: **(a)** Error of Theorem 1 with its own constants, without noise and with noise at the limit of equation 5. **(b)** Lipschitz counter-noise: the threshold in M/g_0 does not depend on d. **(c)** Bounded step noise: thresholds align in the coordinate $\Delta\sqrt{d}$, i.e. scale as $1/\sqrt{d}$.

Bias versus dimension (Figure 1, left). For $f \equiv 0$ and noise depending on one coordinate, the mean of the estimator reduces to a one-dimensional integral with density $\propto (1-t^2)^{(d-3)/2}$, which we compute without Monte Carlo error. The bias of a step matches $\frac{d\Delta}{\gamma}\mathbb{E}|t|$ to 10^{-4}, growing from 0.034 ($d = 4$) to 0.511 ($d = 1024$); the bias of an M-Lipschitz ramp never exceeds M.

The theorems at their own constants (Figure 2). With the parameters of Theorem 1 ($\varepsilon = 0.2$, $D = 2$, $d \in \{5, 10, 20, 40\}$, N up to $1.9 \cdot 10^5$) and noise at the limit of equation 5, the worst error over 8 runs is at most 0.083ε: the constants are conservative by about $12\times$. Against the "counter" noise of Theorem 2 on a linear objective with gradient g_0, the error as a function of M/g_0 is the same for $d = 5, 20, 80$, whereas the failure threshold of a bounded step noise moves as $1/\sqrt{d}$.

Floating-point rounding (Figure 3). We evaluate $f(x) = 1 + \|x - c\|_1/\sqrt{d}$ ($M_f = 1$) in fp64, fp32, bf16, fp16, and fp16 with unbiased stochastic rounding. Rounding is far from worst-case: the bound $M_f\gamma + D\sqrt{d}\,q/(2\gamma)$ is pessimistic by two to three orders of magnitude. For small γ the method fails for a different reason: the two rounded values coincide, the estimate is zero, and the iterate does not move. The threshold for bf16 is $8 = 2^3$ times that for fp16, exactly the difference in mantissa bits, and stochastic rounding moves it down by about $3\times$. Measuring the threshold in fp16 for $d = 4, \ldots, 128$, we find $\gamma_{\text{crit}} \approx 0.2\sqrt{d}\,q/M_f$ (slope 0.49 in log–log coordinates, predicted 0.5), in line with the heuristic $|f(x+\gamma e) - f(x-\gamma e)| \approx 2\gamma|\langle \nabla f, e\rangle| \sim 2\gamma M_f/\sqrt{d}$.

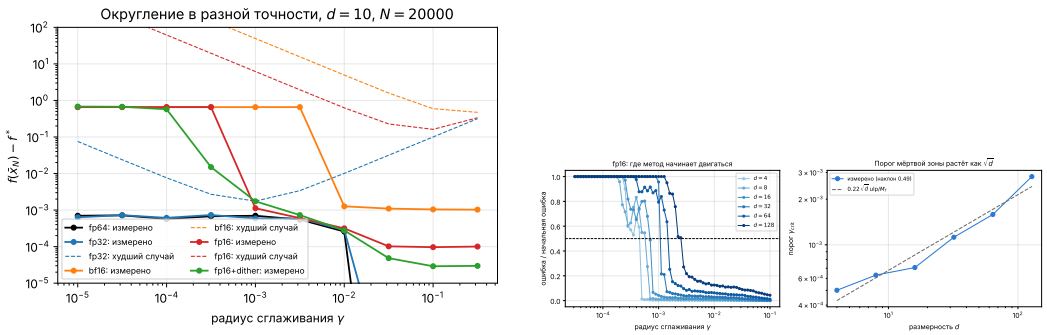


Figure 3: **Left:** final error versus smoothing radius for different precisions ($d = 10$); dashed: worst-case bound. **Right:** dead-zone threshold γ_{crit} in fp16 versus d; the fitted slope is 0.49.

6 Discussion and Limitations

Scope. Our method is non-accelerated; the batched accelerated scheme of Gasnikov et al. (2022b) achieves $O(d^{1/4}M_f D/\varepsilon)$ rounds, and whether the threshold $\Theta(\varepsilon/D)$ survives acceleration requires a bias-robustness argument for accelerated methods with bounded iterates, which we do not provide (cf. Devolder et al., 2014). The dead-zone law is empirical.

Open questions. (i) The motivating example of the conjecture in Dvinskikh et al. (2022, Appendix B) is the error of a sample average, which is Lipschitz with constant $M/\sqrt{n}$ on the queried points but only $\sqrt{d}M/\sqrt{n}$ globally. Whether $\sqrt{d}$ is necessary for noise that is Lipschitz *only on the queried points* appears to be open. (ii) A theory of rounding that explains both the dead zone and the gap to the worst case. (iii) Methods that are simultaneously optimal in calls, rounds and admissible Lipschitz noise.

AI use statement

This work was carried out with substantial help of a generative AI assistant (a large language model with code execution). The assistant was used for literature search and reading, for drafting and checking proofs, for writing the experiment code and tests, and for drafting the text of this paper. Every lemma is additionally checked numerically by unit tests included in the supplementary material, and the novelty claims were checked against the full texts of the cited papers. **[Authors: before submission, state here which parts you verified yourselves (e.g., proofs checked line by line, code reviewed) — do not submit this sentence.]** We take full responsibility for the content of this work.

Reproducibility statement

All assumptions are stated in Section 3; complete proofs are given in Appendix A. The supplementary material contains the code, a single script that reproduces all figures and tables with fixed random seeds, and the unit tests; experimental parameters are listed in Appendix B.

References

Anastasia Bayandina, Alexander Gasnikov, Fariman Guliev, and Anastasia Lagunovskaya. Gradient-free two-point methods for solving stochastic nonsmooth convex optimization problems with small non-random noises. *Automation and Remote Control*, 79(8):1399–1408, 2018.

Olivier Devolder, François Glineur, and Yurii Nesterov. First-order methods of smooth convex optimization with inexact oracle. *Mathematical Programming*, 146(1):37–75, 2014.

John C. Duchi, Michael I. Jordan, Martin J. Wainwright, and Andre Wibisono. Optimal rates for zero-order convex optimization: The power of two function evaluations. *IEEE Transactions on Information Theory*, 61(5):2788–2806, 2015.

Darina Dvinskikh, Vladislav Tominin, Yaroslav Tominin, and Alexander Gasnikov. Gradient-free optimization for non-smooth saddle point problems under adversarial noise. *arXiv preprint arXiv:2202.06114*, 2022. Version 3, 2023.

Abraham D. Flaxman, Adam Tauman Kalai, and H. Brendan McMahan. Online convex optimization in the bandit setting: gradient descent without a gradient. In *Proceedings of the Sixteenth Annual ACM-SIAM Symposium on Discrete Algorithms*, pp. 385–394, 2005.

Alexander Gasnikov, Pavel Dvurechensky, and Yurii Nesterov. Stochastic gradient methods with inexact oracle. *Proceedings of MIPT*, 8(1):41–91, 2016. In Russian. arXiv:1411.4218.

Alexander Gasnikov, Darina Dvinskikh, Pavel Dvurechensky, Eduard Gorbunov, Aleksandr Beznosikov, and Alexander Lobanov. Randomized gradient-free methods in convex optimization. *arXiv preprint arXiv:2211.13566*, 2022a.

Alexander Gasnikov, Anton Novitskii, Vasilii Novitskii, Farshed Abdukhakimov, Dmitry Kamzolov, Aleksandr Beznosikov, Martin Takáč, Pavel Dvurechensky, and Bin Gu. The power of first-order smooth optimization for black-box non-smooth problems. In *International Conference on Machine Learning*, pp. 7241–7265, 2022b.

Guanghui Lan. An optimal method for stochastic composite optimization. *Mathematical Programming*, 133(1):365–397, 2012.

Michel Ledoux. *The Concentration of Measure Phenomenon*, volume 89 of *Mathematical Surveys and Monographs*. American Mathematical Society, 2001.

Aleksandr Lobanov and Alexander Gasnikov. Improved maximum noise level estimation in black-box optimization problems. *Zapiski Nauchnykh Seminarov POMI*, 540:132–147, 2024.

Timofei Loginov, Darina Dvinskikh, Nazarii Tupitsa, Osman Osmanov, Yuriy Dorn, and Alexander Gasnikov. Minimax oracle call complexity, rounds and admissible noise in zeroth-order optimization of highly smooth functions. *arXiv preprint arXiv:2610.05517*, 2026.

Sadhika Malladi, Tianyu Gao, Eshaan Nichani, Alex Damian, Jason D. Lee, Danqi Chen, and Sanjeev Arora. Fine-tuning language models with just forward passes. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Arkadi S. Nemirovsky and David B. Yudin. *Problem Complexity and Method Efficiency in Optimization*. Wiley, 1983.

Yurii Nesterov and Vladimir Spokoiny. Random gradient-free minimization of convex functions. *Foundations of Computational Mathematics*, 17(2):527–566, 2017.

Dmitry Pasechnyuk-Vilensky, Igor Pavlov, Martin Takáč, and Alexander Gasnikov. Information-theoretic upper bounds for deterministic noise in zeroth-order convex optimization. *arXiv preprint arXiv:2306.16371*, 2023.

Andrej Risteski and Yuanzhi Li. Algorithms and matching lower bounds for approximately-convex optimization. In *Advances in Neural Information Processing Systems*, volume 29, 2016.

Ohad Shamir. An optimal algorithm for bandit and zero-order convex optimization with two-point feedback. *Journal of Machine Learning Research*, 18(52):1–11, 2017.

A PROOFS

Throughout, $\mathcal{F}_k = \sigma(e_0, \dots, e_{k-1})$; the point x_k is $\mathcal{F}_k$-measurable and e_k is independent of $\mathcal{F}_k$.

Fact A (smoothing; Flaxman et al., 2005; Gasnikov et al., 2022a)**.** *If h is continuous, then $h_\gamma(x) = \mathbb{E}_u h(x+\gamma u)$ is differentiable and $\nabla h_\gamma(x) = \frac{d}{\gamma}\mathbb{E}_e[h(x+\gamma e)e] = \frac{d}{2\gamma}\mathbb{E}_e[(h(x+\gamma e) - h(x-\gamma e))e]$. If h is M-Lipschitz, so is h_γ, hence $\|\nabla h_\gamma\| \le M$. If h is moreover convex, h_γ is convex and $h \le h_\gamma \le h + \gamma M$.*

The first identity is the divergence theorem on the ball of radius γ (the ratio of the area of the sphere to the volume of the ball is d/γ); the second follows from the symmetry $e \mapsto -e$.

Fact B (sphere; Ledoux, 2001)**.** *Let $e \sim \mathrm{Unif}(S^{d-1})$, $d \ge 2$. (a) $\mathbb{E}\langle e, r\rangle^2 = \|r\|^2/d$, hence $\mathbb{E}|\langle e, r\rangle| \le \|r\|/\sqrt{d}$. (b) If φ is L_φ-Lipschitz on S^{d-1}, then $\operatorname{Var}\varphi(e) \le L_\varphi^2/(d-1)$.*

Lemma 3 (projected SGD with biased gradients)**.** *Let F be convex and differentiable, $x_F^* \in \arg\min_Q F$, $x_0 \in Q$, and let g_k be $\mathcal{F}_{k+1}$-measurable with $\mathbb{E}[g_k \mid \mathcal{F}_k] = \nabla F(x_k) + b_k$, $\|b_k\| \le B$ and $\mathbb{E}[\|g_k\|^2 \mid \mathcal{F}_k] \le G^2$ almost surely. Then for $x_{k+1} = \Pi_Q(x_k - hg_k)$, $\mathbb{E}F(\bar{x}_N) - F(x_F^*) \le \frac{\|x_0 - x_F^*\|^2}{2hN} + \frac{hG^2}{2} + BD$.*

Proof. Let $r_k = \|x_k - x_F^*\|^2$. Since projection onto Q is non-expansive and $x_F^* \in Q$, $r_{k+1} \le r_k - 2h\langle g_k, x_k - x_F^*\rangle + h^2\|g_k\|^2$. Taking $\mathbb{E}[\cdot \mid \mathcal{F}_k]$ gives $\mathbb{E}[r_{k+1} \mid \mathcal{F}_k] \le r_k - 2h\langle \nabla F(x_k) + b_k, x_k - x_F^*\rangle + h^2G^2$. By convexity $\langle \nabla F(x_k), x_k - x_F^*\rangle \ge F(x_k) - F(x_F^*)$, and $|\langle b_k, x_k - x_F^*\rangle| \le BD$ because $x_k, x_F^* \in Q$. Taking full expectations, summing over $k < N$ (the sum telescopes and is at most r_0) and dividing by $2hN$ bounds the average of $\mathbb{E}F(x_k) - F(x_F^*)$; Jensen's inequality $F(\bar{x}_N) \le \frac{1}{N}\sum_k F(x_k)$ concludes. □

Proof of Lemma 1. Taking $y = x$ gives $\delta_L \le \delta$. By Definition 1, $\delta(y) + M\|x - y\| \ge \delta(x) - \Delta_0$ for all y, so $\delta_L \ge \delta - \Delta_0 > -\infty$. An infimum of M-Lipschitz functions that is finite at one point is finite everywhere and M-Lipschitz. Conversely, if $\delta = \ell + \beta$ with ℓ M-Lipschitz and $\beta \in [0, \Delta_0]$, then $|\delta(x) - \delta(y)| \le M\|x - y\| + |\beta(x) - \beta(y)| \le M\|x - y\| + \Delta_0$. □

Proof of Lemma 2. Write $\delta = \delta_L + \delta_B$ by Lemma 1; δ_L is continuous and $\delta_B = \delta - \delta_L$ is Borel with values in $[0, \Delta_0]$. Accordingly $g = g_1 + g_2$, where g_1 is built from the continuous function $f + \delta_L$ and g_2 from δ_B.

Bias. By Fact A applied to f and to δ_L, $\mathbb{E}g_1 = \nabla f_\gamma(x) + \nabla(\delta_L)_\gamma(x)$ with $\|\nabla(\delta_L)_\gamma(x)\| \le M$. For a unit r, since $|\delta_B(a) - \delta_B(b)| \le \Delta_0$, Fact B(a) gives $|\langle \mathbb{E}g_2, r\rangle| \le \frac{d}{2\gamma}\Delta_0\mathbb{E}|\langle e, r\rangle| \le \frac{\sqrt{d}\Delta_0}{2\gamma}$. Taking the supremum over r and setting $b = \nabla(\delta_L)_\gamma(x) + \mathbb{E}g_2$ gives the bias bound.

Second moment. $\|g\|^2 \le 2\|g_1\|^2 + 2\|g_2\|^2$. The function $\varphi(e) = (f + \delta_L)(x + \gamma e)$ is $\gamma(M_f + M)$-Lipschitz on the sphere, $\|g_1\|^2 = \frac{d^2}{4\gamma^2}(\varphi(e) - \varphi(-e))^2$, and $(\varphi(e) - \varphi(-e))^2 \le 2(\varphi(e) - \mathbb{E}\varphi)^2 + 2(\varphi(-e) - \mathbb{E}\varphi)^2$. Since $-e$ has the same law as e, Fact B(b) yields $\mathbb{E}\|g_1\|^2 \le \frac{d^2}{4\gamma^2} \cdot \frac{4\gamma^2(M_f+M)^2}{d-1} \le 2d(M_f + M)^2$. Finally $\|g_2\|^2 \le \frac{d^2\Delta_0^2}{4\gamma^2}$. □

Proof of Theorem 1. Conditional moments. Since e_k is independent of $\mathcal{F}_k$, x_k is $\mathcal{F}_k$-measurable and δ is fixed, the conditional law of g_k given $\mathcal{F}_k$ is the law of the estimator of Lemma 2 at x_k. Hence $\mathbb{E}[g_k \mid \mathcal{F}_k] = \nabla f_\gamma(x_k) + b(x_k)$ with $\|b(x_k)\| \le \frac{\varepsilon}{6D} + \frac{\sqrt{d}}{2} \cdot \frac{3M_f}{\varepsilon} \cdot \frac{\varepsilon^2}{9\sqrt{d}M_f D} = \frac{\varepsilon}{3D} =: B$. Similarly $\frac{d^2\Delta_0^2}{2\gamma^2} = \frac{d\varepsilon^2}{18D^2} \le \frac{dM_f^2}{18} \le d(M_f + M)^2$ (using $\varepsilon \le M_f D$), so $\mathbb{E}[\|g_k\|^2 \mid \mathcal{F}_k] \le 5d(M_f + M)^2 = G^2$.

SGD. With $F = f_\gamma$, $\|x_0 - x_F^*\| \le D$ and $h = D/(G\sqrt{N})$, Lemma 3 gives $\mathbb{E}f_\gamma(\bar{x}_N) - \min_Q f_\gamma \le \frac{DG}{\sqrt{N}} + BD \le \frac{\varepsilon}{3} + \frac{\varepsilon}{3}$, because $N \ge 45d(M_f + M)^2D^2/\varepsilon^2 = 9D^2G^2/\varepsilon^2$.

Back to f. By Fact A, $f \le f_\gamma$ and $\min_Q f_\gamma \le f_\gamma(x^*) \le f^* + \gamma M_f = f^* + \varepsilon/3$. Adding up gives $\mathbb{E}f(\bar{x}_N) - f^* \le \varepsilon$. □

Proof of Theorem 2, details. The interval $(2\varepsilon/D, \min\{M, M_f\}]$ is nonempty because $M > 2\varepsilon/D$ and $M_f > 2\varepsilon/D$. The functions $f_\pm$ are convex and a-Lipschitz with $a \le M_f$, and $\delta_\pm$ are a-Lipschitz with $a \le M$, so $\delta_\pm \in \mathcal{N}(M, 0)$. As the oracle returns $f_\pm + \delta_\pm \equiv 0$, the joint law of all queries, answers and the output is the same under both instances, and one of the two expected errors exceeds ε. □

Proof of Proposition 1. (a) The proof of Theorem 1 used only that, conditionally on $\mathcal{F}_k$, the noise is a fixed function of class $\mathcal{N}(M, \Delta_0)$ independent of e_k; Lemma 2 applies conditionally (joint measurability allows freezing $\mathcal{F}_k$). (b) δ_k is M-Lipschitz, and at $x_k \pm \gamma e_k$ it equals $\pm\gamma M \operatorname{sign}\langle e_k, v\rangle$. Hence the noise part of the estimator is $dM \operatorname{sign}\langle e_k, v\rangle\, e_k$, whose mean is $dM\, \mathbb{E}|\langle e, v\rangle|\, v$ by symmetry, and $d\,\mathbb{E}|\langle e, v\rangle| = d\,\Gamma(d/2)/(\sqrt{\pi}\,\Gamma((d+1)/2)) \sim \sqrt{2d/\pi}$. □

B Experimental details

Bias versus dimension. $f \equiv 0$; noise $\delta(x) = \varphi(x_1)$ with a step $\varphi(z) = \Delta \operatorname{sign} z$ or a ramp $\varphi(z) = \Delta \operatorname{clip}(Mz/\Delta, -1, 1)$; $\Delta = 10^{-3}$, $\gamma = 0.05$, $M \in \{0.01, 0.03, 0.1, 0.3\}$, d from 4 to 1024; maximum over the offset $s \in [-2\gamma, 2\gamma]$ of the point from the jump. The mean of the estimator is a one-dimensional integral over $t = \langle e, e_1\rangle$ with density $\propto (1 - t^2)^{(d-3)/2}$, computed on a grid of $4 \cdot 10^5$ points.

Phase diagram. $f(x) = -g_0\langle u, x\rangle$ on the unit ball with $g_0 = 2\varepsilon$, $\varepsilon = 0.1$; noise: a ramp of amplitude Δ and slope M across $\langle u, x\rangle = 0$ (a step for $M = \infty$); start $x_0 = -0.6u$, $d = 20$, $\gamma = 0.05$, 3000 iterations, 16 runs per cell on a 13×13 grid. Disagreements with the prediction occur only next to the boundary $M \approx g_0$, where the total slope is small and the method is slowed down rather than stopped; with 10^4 iterations all runs in that column succeed.

Theorems. (a) $\varepsilon = 0.2$, unit ball ($D = 2$), $M_f = 1$, parameters exactly as in Theorem 1; objectives $a\|x - c\|$ with $\|c\| = 0.5$ and $a \in \{M_f, 3\varepsilon/D\}$; noise $M\langle u, x\rangle + \Delta_0 \mathbb{1}[\langle u, x\rangle > 0]$ at the limit of equation 5; $d \in \{5, 10, 20, 40\}$, N from 23,257 to 186,050, 8 runs. (b) Linear objective as in the phase diagram, noise $M\langle u, x\rangle$, $M/g_0 \in [0, 2]$, $d \in \{5, 20, 80\}$, 4000 iterations. (c) Same objective, step noise $\Delta \operatorname{sign}\langle u, x\rangle$ with 22 values of Δ in $[10^{-4}, 0.3]$.

Rounding. $f(x) = 1 + \|x - c\|_1/\sqrt{d}$, $d = 10$, ball of radius 2, $2 \cdot 10^4$ iterations, 6 runs, 10 values of γ in $[10^{-5}, 0.3]$. bf16 is emulated by rounding fp32 results to 8 mantissa bits; stochastic rounding rounds the exact value up or down to the fp16 grid without bias. Dead zone: fp16, $q = 2^{-10}$, $d \in \{4, \dots, 128\}$, 71 values of γ in $[3 \cdot 10^{-5}, 0.1]$, 3000 iterations, 6 runs; γ_{crit} is the smallest γ for which the method at least halves the initial error. Measured $\gamma_{\mathrm{crit}} M_f/(\sqrt{d}q)$: $0.26, 0.23, 0.18, 0.20, 0.20, 0.26$ for $d = 4, 8, 16, 32, 64, 128$.